PHY · Particle accelerators and mass–energy — syllabus points and past-paper evidence

Every award of the 2016–2025 papers, labelled to the 2026/2027 Year 12 syllabus content (identical in both). Statistics from the labels; notes written by AI from that evidence: review before use.

Pattern 1: Fields controlling charged particles in accelerators

Syllabus: S26-U4-SU-18 / S27-U4-SU-18

Frequency: 10 of 10 exam years; 108 marks in all.

Marks reward: Substitution and working 36, Qualitative explanation 25, Final answer 23, Select the principle or relationship 22, Diagram 2

Past questions: 2016P-17(c); 2017C-21(ai); 2017C-21(ci); 2017C-21(d); 2018C-19(b); 2018C-19(d); 2018C-19(f); 2019C-19(a); 2019C-19(b); 2019C-19(c); 2019C-19(d); 2019C-19(e); 2019P-15(ei); 2019P-15(eii); 2020C-20(c); 2020C-20(di); 2020C-20(dii); 2021R-11(a); 2021R-11(b); 2022R-4(a); 2023C-19(a); 2023C-19(f); 2023P-13(c); 2023P-18(aii)…

Summary: This point examines how electric fields change a charged particle's speed and energy, and how magnetic fields bend its path, in accelerators, cyclotrons, mass spectrometers and velocity selectors. It usually appears as a multi-part question that chains an energy or force equation to a radius, speed, mass or frequency, often followed by a short explanation of what happens when one variable changes.

Recognise: Look for a charged particle (electron, proton, ion, pion) entering a uniform electric field between plates, a magnetic field region, or both together. Key words include accelerating potential, potential difference, kinetic energy gained, circular path, radius, cyclotron, mass spectrometer, velocity selector and detector position. Questions asking why a quantity changes, or why a path is or is not circular, are explanation questions on the same relationships.

Steps

1. Decide which field acts. An electric field between plates changes speed, with $F = qE$, $E = V/d$ and work done $W = qV$. A magnetic field changes only direction, with $F = qvB$ acting at right angles to the velocity.
2. For acceleration through a potential difference, equate energy gained to kinetic energy gained: $qV = \frac{1}{2}mv^2$, so $v = \sqrt{(2qV/m)}$. Convert energies given in eV or larger units to joules before using them.
3. For motion in a uniform magnetic field, equate the magnetic force to the centripetal force: $qvB = mv^2/r$, so $r = mv/(qB) = p/(qB)$, where $p = mv$ is momentum.
4. To find mass or charge from a measured radius, rearrange the same relation, for example $m = qBr/v$. Use the diameter for a beam width and halve it to get the radius.
5. For a velocity selector, balance the forces so the particle passes straight through: $qE = qvB$, giving $v = E/B$.
6. For cyclotron frequency, combine period $T = 2\pi r/v$ with $r = mv/(qB)$ to get $f = qB/(2\pi m)$. This depends on neither speed nor radius, which is why the field can alternate at a fixed frequency.
7. For deflection in a region with only an electric field, treat the horizontal and vertical motions separately and use $a = F/m$ with $s = ut + \frac{1}{2}at^2$ and $v^2 = u^2 + 2as$.

Where the marks are

- Each calculation mark is usually given for stating the correct equation before substituting values, so write the relationship first.
- Answer marks require a correct value with units and sensible significant figures, and a unit error loses the mark even when the method is right.
- Explanation marks reward a chain: name the equation, state how one variable changes, then conclude what happens to the particle's path, speed or frequency.
- Diagram marks reward showing relative sizes correctly, such as a heavier or faster particle with a larger radius, and stating the ratio where a quantitative comparison is asked.

- Principle marks are commonly awarded for equating the magnetic force to centripetal force, or equating electric work to kinetic energy, even before any numbers are used.

Pitfalls

- Treating a magnetic field as if it speeds a particle up. The magnetic force is perpendicular to velocity, so it changes direction but not speed; only the electric field changes kinetic energy.
- Forgetting to convert eV to J (multiply by the elementary charge) or to convert GeV, TeV, cm or mm to base SI units before substituting.
- Using the diameter as the radius, or failing to halve it when a beam width is given.
- Concluding that a path is circular when the magnetic field is not uniform, or ignoring that a changing speed changes the radius.
- Giving explanations that state a result without linking it to the governing equation, so the explanation mark is lost even when the conclusion is right.

Pattern 2: Relativistic momentum

Syllabus: S26-U4-SU-19 / S27-U4-SU-19

Frequency: 5 of 10 exam years; 28 marks in all.

Marks reward: Qualitative explanation 11, Select the principle or relationship 6, Substitution and working 5, Final answer 4, Evaluate and conclude 1, Graph work 1

Past questions: 2017C-21(cii); 2018C-19(c); 2018C-19(f); 2021P-18(b); 2021P-18(c); 2021P-18(d); 2024C-19(d); 2024P-14(a); 2024P-14(b); 2024P-14(c); 2025R-10

Summary: This point examines how relativistic momentum grows as an object's speed approaches c, so the momentum and kinetic energy needed diverge and no object with mass can reach the speed of light. It usually appears as short calculations using p = mv / sqrt(1 - v^2/c^2) in accelerator or collision contexts, and as explain-type questions asking why c cannot be reached.

Recognise: Look for a question that gives a particle's mass and a speed that is a large fraction of c, then asks for its momentum or for its momentum in another frame. Explain questions often refer to a graph of kinetic energy against speed, or ask why an object cannot reach the speed of light. The relativistic momentum equation is in the Formulae and Data booklet, so it must be recalled and applied rather than given.

Steps

1. Identify the rest mass m and the speed v, converting v to m s^-1 if it is given as a fraction of c.
2. Write the relation p = mv / sqrt(1 - v^2/c^2). The factor 1/sqrt(1 - v^2/c^2) is the Lorentz factor gamma, so p = gamma m v.
3. For a question set in another frame, first find the relative speed the question asks for (using relativistic velocity addition if needed), then apply the momentum formula in that frame.
4. Substitute carefully: work out v^2/c^2 (or v/c) before subtracting from 1, and keep extra digits, because 1 - v^2/c^2 becomes very small close to c.
5. For a limit or explanation question, look at the denominator as v approaches c: 1 - v^2/c^2 tends to zero, so p and gamma tend to infinity.
6. Link the divergence to energy: since Ek = (gamma - 1) m c^2, an unbounded gamma means unbounded kinetic energy, so reaching c would need infinite energy.
7. Give the numerical answer with the correct unit (kg m s^-1) and correct significant figures, and finish with a sentence stating the conclusion in words.

Where the marks are

- A mark for writing the correct relativistic momentum relation with the square root over the whole denominator, not p = mv.
- A mark for substituting the correct mass and speed, with the speed expressed consistently in m s^-1 or as a fraction of c.
- A mark for the final numerical answer with correct significant figures and units.

- Explanation questions usually need two linked points: the denominator tends to zero as v approaches c, so p (or Ek) tends to infinity.
- A concluding statement that an object with mass cannot reach c, since this would need infinite momentum or energy. For frame-change questions, a mark is given for finding the relative speed before applying the formula.

Pitfalls

- Using the classical p = mv without the relativistic factor, which underestimates momentum at high speed.
- Writing the denominator incorrectly, such as 1 - v/c or a wrongly squared c, which breaks the limit argument.
- Stating only that mass increases or that c cannot be reached, without linking it to the denominator tending to zero and momentum or energy tending to infinity.
- Rounding 1 - v^2/c^2 too early when v is close to c, which gives a badly wrong answer.
- Using the lab speed directly in another frame instead of the relative speed, or omitting the unit kg m s^-1 (or N s) from the answer.

Pattern 3: Mass–energy equivalence

Syllabus: S26-U4-SU-20 / S27-U4-SU-20

Frequency: 9 of 10 exam years; 50 marks in all.

Marks reward: Substitution and working 19, Select the principle or relationship 12, Qualitative explanation 12, Final answer 6, Recall 1

Past questions: 2017C-21(cii); 2017P-15(aii); 2018C-19(g); 2018C-19(h); 2018P-17(d); 2019R-5; 2020R-9; 2021P-18(a); 2021R-2; 2021R-4; 2022P-14(b); 2022P-14(c); 2023C-19(b); 2024P-14(d); 2025P-15(b); 2025R-10; 2025R-12(b)

Summary: Mass-energy equivalence ($E = mc^2$ for rest energy, and the total energy of a moving object $E = mc^2/\sqrt{(1 - v^2/c^2)}$) links an object's mass with its energy and explains where energy comes from in nuclear and particle reactions. Papers usually test it with calculations that find rest energy, speed or kinetic energy from a given energy, and with short explanations of why a massive object cannot reach the speed of light or why reaction products can have more mass than the reactants.

Recognise: Look for energy given in J, eV, keV, MeV, GeV or TeV alongside a mass, a speed written as a fraction of c, or a reaction where particles are created, annihilated or collide at high speed. Phrases such as 'rest energy', 'total energy', 'kinetic energy of the particle', 'show that the mass is', 'calculate the speed in terms of c' or 'why can nothing with mass reach the speed of light' signal this point.

Steps

1. Decide which energy the question means: rest energy $E0 = mc^2$ for a stationary object; total energy $E = \gamma mc^2$ for a moving one, where $\gamma = 1/\sqrt{(1 - v^2/c^2)}$; kinetic energy $Ek = E - mc^2$.
2. Convert all energies to joules and all masses to kilograms before substituting ($1 \text{ eV} = 1.60 \times 10^{-19} \text{ J}$), and check the powers of ten for keV, MeV, GeV and TeV.
3. To find a mass from a known rest energy, rearrange $m = E/c^2$. For a moving particle, first find γ from the given speed, then use $E = \gamma mc^2$.
4. To find a speed from a known total energy, write $\sqrt{(1 - v^2/c^2)} = mc^2/E$, square both sides and rearrange to $v = c\sqrt{(1 - (mc^2/E)^2)}$. Give the result in the form the question asks, such as a multiple of c or in m s^{-1}.
5. For a reaction, apply conservation of total energy: the total energy before equals the total energy after, with each particle's energy including its rest energy plus kinetic energy. Any rise in the rest mass of the products must come from kinetic energy of the incoming particles or from photon energy.
6. For pair production or annihilation, use the threshold $hf = 2mc^2$ for a photon creating a particle-antiparticle pair. For a particle and antiparticle annihilating at rest, the two photons leave in opposite directions and each carries energy mc^2.
7. Check the answer: a speed must be less than c, mass values should be very small for everyday

energies, and the number of significant figures should match the question.

Where the marks are

- Award for choosing the correct form of energy (rest, total or kinetic) and writing the matching equation.
- Award for converting units to J and kg correctly and keeping c^2 in the working.
- Award for an algebra step that isolates the speed or mass before any numbers are substituted.
- Award for a final answer with correct units and significant figures, given as a multiple of c when the question asks for it.
- Explanation marks come from stating that as v approaches c, γ and the energy grow without limit so infinite energy would be needed, and that products are heavier because kinetic energy is converted into rest mass.

Pitfalls

- Using $E = mc^2$ for a moving particle instead of the total relativistic energy, or forgetting to subtract the rest energy when the question asks for kinetic energy.
- Saying the mass increases with speed. Rest mass is constant; it is the energy and momentum that increase.
- Unit errors with eV, keV, MeV or GeV, or mixing up grams and kilograms.
- Dropping the square on c, or rounding intermediate values too early so the final speed is wrong in its last figures.
- Giving a speed as a bare number with no units, or without expressing it in terms of c when the question asks for that.

Pattern 4: Total energy: rest energy plus kinetic energy

Syllabus: S26-U4-SU-21 / S27-U4-SU-21

Frequency: 5 of 10 exam years; 9 marks in all.

Marks reward: Select the principle or relationship 5, Substitution and working 3, Final answer 1

Past questions: 2018C-19(h); 2021R-2; 2022P-14(b); 2024P-14(d); 2025P-15(b); 2025R-10; 2025R-12(b)

Summary: This point examines how the total energy of a moving object splits into rest energy (mc^2) and kinetic energy (E minus E_rest). It usually appears inside multi-step questions on particle accelerators, collisions and mass-energy. In past papers it mostly carries principle marks for setting up the energy balance, followed by calculation and final-answer marks.

Recognise: Look for a question giving a rest mass or rest energy (often in MeV, GeV or kg) together with a kinetic energy, total energy or speed of a particle. Also look for questions asking why a massive object cannot reach the speed of light, and collision or decay questions that say total energy is conserved or ask for the kinetic energy shared among products.

Steps

1. Write the rest energy as E_rest = mc^2, using the rest mass in kg, or use a rest energy that is given directly.
2. Convert every energy to one unit before combining. For energies in eV, MeV or GeV, multiply by 1.60 x 10^-19 J per eV and keep the powers of ten straight.
3. Form the total energy as E = E_rest + E_k, adding the rest energy to the kinetic energy given.
4. If kinetic energy is asked for, rearrange to E_k = E - E_rest.
5. If speed is asked for, use E = gamma x E_rest with gamma = 1/sqrt(1 - v^2/c^2), which gives v = c x sqrt(1 - (E_rest/E)^2).
6. For a conservation question, set total energy before equal to total energy after, or set the mass energy released equal to the change in total kinetic energy, delta(mc^2) = delta(sum of E_k).
7. Check the answer: speed must be less than c, and the unit (kg, J, keV or m/s) must match the question.

Where the marks are

- A mark for stating total energy as rest energy plus kinetic energy before any numbers are substituted.
- A mark for identifying rest energy as mc^2 and using the mass in consistent units.
- A mark for converting MeV, GeV or eV into joules, or keeping units consistent, before calculating.
- A mark for the conservation step in collision or decay questions, equating energy before and after.
- A mark for rearranging the formula correctly when mass or speed is the unknown, then a mark for the final answer with units and sensible significant figures.

Pitfalls

- Forgetting to add the rest energy to the kinetic energy, or subtracting in the wrong direction.
- Leaving energies in eV, MeV or GeV while using mc^2 in SI units, which gives answers out by a large power of ten.
- Using the classical kinetic energy formula (one half m v^2) at speeds close to c, where the relativistic formula is required.
- Mixing keV and J, or giving a kinetic energy in the wrong unit for the question.
- Rounding intermediate values too early, which loses the final mark for significant figures.

Pattern 5: Energy–momentum relation

Syllabus: S26-U4-SU-22 / S27-U4-SU-22

Frequency: 0 of 10 exam years; 0 marks in all.

Summary: This point asks you to link an object's total energy to its mass and momentum through E^2 = (pc)^2 + (mc^2)^2, and to use it for massive particles, photons and rest-energy or kinetic-energy calculations. No past-paper awards exist for this point (parts_total 0), so this note is written from the physics alone and the marks and pitfalls are generic expectations rather than examiner-derived.

Recognise: Look for questions that give a particle's momentum and mass (or energy) and ask for total energy, kinetic energy or speed, or that describe a photon or a particle moving at close to light speed in an accelerator. Any mention of total energy, rest energy, relativistic momentum or the term mc^2 signals this relationship.

Steps

1. Write down the known quantities and check units: momentum in kg m s^-1 or MeV/c, mass in kg or MeV/c^2, energy in J or MeV, and c = 3.00 x 10^8 m s^-1.
2. State the energy-momentum relation E^2 = (pc)^2 + (mc^2)^2 and identify which terms apply to the object (a photon has m = 0).
3. Substitute the values, keeping the powers of c correct, then take the positive square root to get E.
4. To find momentum instead, rearrange: p = sqrt(E^2 - (mc^2)^2) / c.
5. For kinetic energy, use K = E - mc^2 (E minus rest energy), not E alone.
6. Check the answer: total energy must be at least the rest energy mc^2, and for a photon E must equal pc.

Where the marks are

- A mark for stating the energy-momentum relation before substituting values.
- A mark for correctly identifying which term vanishes (for example m = 0 for a photon, or p = 0 for a particle at rest).
- A mark for correct unit handling and powers of c in the substitution.
- A mark for the final rearranged or subtracted result, such as kinetic energy as E minus mc^2.
- A mark for a sensible check, such as E being no smaller than mc^2.

Pitfalls

- Using E = pc for a massive particle, which only holds for massless particles.

- Forgetting to square the terms or dropping the c powers, so the units do not balance.
- Reporting total energy when the question asks for kinetic energy, or subtracting the wrong rest energy.
- Mixing MeV and J, or MeV/c with MeV, without converting.
- Taking the negative root of the square, which gives a physically meaningless negative energy.